

Surfaces of constant principal– curvatures ratio in isotropic geometry: auxiliary computations.

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Here we check auxiliary computations in the proofs.

1. Checking of Theorem 21

Isotropic Gaussian Curvature

$$K[u_] := (u^3 * f'[u] * f'[u] - 1) / u^4$$

$$K[u] = \frac{-1 + u^3 f'[u] f''[u]}{u^4}$$

Isotropic Mean Curvature

$$H[u_] := (f'[u] + u * f''[u]) / (2 * u)$$

$$H[u] = \frac{f'[u] + u f''[u]}{2 u}$$

The equation $H^2 / K - (a + 1)^2 / (4 a) = 0$

$$eq := H[u]^2 / K[u] - (a + 1)^2 / (4 * a)$$

eq

$$-\frac{(1+a)^2}{4a} + \frac{u^2 (f'[u] + u f''[u])^2}{4(-1+u^3 f'[u] f''[u])}$$

1.1. Checking that equations (13) and (14) are equivalent

eq1 :=

$$((a-1) * ((u^2 * f''[u] + u * f'[u]) / (2 * (a+1))))^2 - ((u^2 * f''[u] - u * f'[u]) / 2)^2 - 1$$

eq1

$$-1 - \frac{1}{4} (-u f'[u] + u^2 f''[u])^2 + \frac{(-1+a)^2 (u f'[u] + u^2 f''[u])^2}{4(1+a)^2}$$

FullSimplify[eq / eq1]

$$\frac{1+2a+a^2}{4a-4au^3 f'[u] f''[u]}$$

1.2. Checking of equations (15) and (16)

$$\text{FullSimplify}\left[\text{Solve}\left[\text{Csc}[2 * s[u]] == \frac{(-1+a)(u f'[u] + u^2 f''[u])}{2 * (1+a)} \&\&\right.\right.$$

$$\left.\left.\text{Cot}[2 * s[u]] == \frac{1}{2} (u f'[u] - u^2 f''[u]), \{f'[u], f''[u]\}\right]\right]$$

$$\left\{\left\{f'[u] \rightarrow \frac{a \text{Cot}[s[u]] + \text{Tan}[s[u]]}{(-1+a)u}, f''[u] \rightarrow \frac{\text{Cot}[s[u]] + a \text{Tan}[s[u]]}{(-1+a)u^2}\right\}\right\}$$

1.3. Checking of equation (17)

$$\text{FullSimplify}\left[\text{Solve}\left[\partial_u \left(\frac{a \text{Cot}[s[u]] + \text{Tan}[s[u]]}{(-1+a)u}\right) == \frac{\text{Cot}[s[u]] + a \text{Tan}[s[u]]}{(-1+a)u^2}, \{s'[u]\}\right]\right]$$

$$\left\{\left\{s'[u] \rightarrow \frac{2(1+a) \text{Csc}[2s[u]]}{u(-a \text{Csc}[s[u]]^2 + \text{Sec}[s[u]]^2)}\right\}\right\}$$

$$\text{FullSimplify}\left[\frac{2(1+a) \text{Csc}[2s[u]]}{u(-a \text{Csc}[s[u]]^2 + \text{Sec}[s[u]]^2)} * (\text{Tan}[s[u]] - a * \text{Cot}[s[u]]) / (a+1)\right]$$

$$\frac{1}{u}$$

1.4. Checking of equation (18)

```
In[2]:= f[s_] := s + Cot[2 * s] + (a^2 + 1) Csc[2 * s] / (a^2 - 1) + c3
In[3]:= Simplify[D_s (f[s]) - (Tan[s] + a * Cot[s]) * (Tan[s] - a * Cot[s]) / ((a - 1) * (a + 1))]
Out[3]= 0
```

1.4.1. Another representation of the equation(18)

```
In[4]:= f1[s_] := s + Tan[s] / (a^2 - 1) + a^2 * Cot[s] / (a^2 - 1) + c3
In[5]:= Simplify[D_s (f[s]) - (Tan[s] + a * Cot[s]) * (Tan[s] - a * Cot[s]) / ((a - 1) * (a + 1))]
Out[5]= 0
```

2. Checking of Lemma 25

Isotropic Gaussian curvature

```
gp (v) = g' (v)
gpp (v) = g'' (v)
b := (a + 1) / (a - 1)
K[u_, v_] := f' [u] * f'' [u] * gpp[v]
K[u, v]
gpp[v] f' [u] f'' [u]
```

Isotropic Mean curvature

```
H[u_, v_] := -(f' [u] * gpp[v] + (1 + gp[v] * gp[v]) * f'' [u]) / 2
H[u, v]
1
2 (-gpp[v] f' [u] - (1 + gp[v]^2) f'' [u])
```

2.1 Checking of equation (29)

```
f[u_] := Exp[λ * u]
gpp := λ * ((a + 1) + Sqrt[(a - 1)^2 - 4 * a * g' [v]^2])^2 / (4 * a)
gpp
λ (1 + a + sqrt((-1 + a)^2 - 4 a g' [v]^2))^2
4 a
```

```
FullSimplify[( (1 + g'[v]^2) * f''[u] / f'[u] + gpp)^2 - (a + 1)^2 * f''[u] * gpp / (a * f'[u])]
0
```

2.2 Checking of equations (32) and (33)

$$\frac{\text{FullSimplify}\left[\partial_p \left(-\text{Log}\left[a + 1 + \sqrt{(a-1)^2 - 4ap^2}\right] - \frac{a+1}{a+1 + \sqrt{(a-1)^2 - 4ap^2}} + C1 \right)\right]}{(1+a + \sqrt{(-1+a)^2 - 4ap^2})^2}$$

$$\frac{\text{FullSimplify}\left[\partial_p \left(-\text{Log}\left[a + 1 - \sqrt{(a-1)^2 - 4ap^2}\right] - \frac{a+1}{a+1 - \sqrt{(a-1)^2 - 4ap^2}} + C1 \right)\right]}{(1+a - \sqrt{(-1+a)^2 - 4ap^2})^2}$$

$$\frac{\text{FullSimplify}\left[\partial_p \left(\frac{(-1+a)^2 \left(\text{ArcTan}[p] + \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2 - 4ap^2}}\right] \right)}{4a} + \frac{p(1+a)}{a+1 - \sqrt{(a-1)^2 - 4ap^2}} + C2 \right)\right]}{(1+a - \sqrt{(-1+a)^2 - 4ap^2})^2}$$

$$\frac{\text{FullSimplify}\left[\partial_p \left(\frac{(-1+a)^2 \left(\text{ArcTan}[p] - \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2 - 4ap^2}}\right] \right)}{4a} + \frac{p(1+a)}{a+1 + \sqrt{(a-1)^2 - 4ap^2}} + C2 \right)\right]}{(1+a + \sqrt{(-1+a)^2 - 4ap^2})^2}$$

2.3 Checking of equations (34) and (35)

Case “-“ between tangents.

$$b := (a + 1) / (a - 1)$$

$$w := \text{ArcTan}[p] - \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2 - 4ap^2}}\right] + \text{Pi} / 2 * \text{Sign}[a - 1]$$

When $a > 1$

$$w := \text{ArcTan}[p] - \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2 - 4ap^2}}\right] + \text{Pi} / 2$$

Finding $\sin(w)$

FullSimplify[**TrigExpand**[**Sin**[**w**]]]

$$\frac{(1+a)p^2 + \sqrt{(-1+a)^2 - 4ap^2}}{\sqrt{1+p^2} \sqrt{\frac{(-1+a)^2(1+p^2)}{(-1+a)^2 - 4ap^2}} \sqrt{(-1+a)^2 - 4ap^2}}$$

Finding $\cos(w)$

FullSimplify[**TrigExpand**[**Cos**[**w**]]]

$$\frac{p(1+a - \sqrt{(-1+a)^2 - 4ap^2})}{\sqrt{1+p^2} \sqrt{\frac{(-1+a)^2(1+p^2)}{(-1+a)^2 - 4ap^2}} \sqrt{(-1+a)^2 - 4ap^2}}$$

$$\sin w := \left((1+a)p^2 + \sqrt{(-1+a)^2 - 4ap^2} \right) / ((a-1) * (1+p^2))$$

$$\cos w := \left(p(1+a - \sqrt{(-1+a)^2 - 4ap^2}) \right) / ((a-1) * (1+p^2))$$

Checking (34)

$$\begin{aligned} & \text{FullSimplify}\left[-\text{Log}\left[\left(1+a + \sqrt{(-1+a)^2 - 4ap^2}\right)\right] - \right. \\ & \quad \left. (a+1) / \left(1+a + \sqrt{(-1+a)^2 - 4ap^2}\right) - ((b^2-1) * \text{Log}[b - \sin w] + b * \sin w) / (b^2-1)\right] \\ & - \frac{(1+a)^2}{4a} - \text{Log}\left[\frac{1+a - \sqrt{(-1+a)^2 - 4ap^2}}{(-1+a)(1+p^2)}\right] - \text{Log}\left[1+a + \sqrt{(-1+a)^2 - 4ap^2}\right] \end{aligned}$$

$$\begin{aligned} & \text{FullSimplify}\left[\left(1+a + \sqrt{(-1+a)^2 - 4ap^2}\right) * \left(1+a - \sqrt{(-1+a)^2 - 4ap^2}\right) / ((-1+a)(1+p^2))\right] \\ & \frac{4a}{-1+a} \end{aligned}$$

Thus it is

$$\begin{aligned} & - \frac{(1+a)^2}{4a} - \text{Log}\left[\frac{4a}{-1+a}\right] \\ & - \frac{(1+a)^2}{4a} - \text{Log}\left[\frac{4a}{-1+a}\right] \end{aligned}$$

Checking (35)

$$\text{FullSimplify}\left[(a-1)^2 / (4a) * \left(\text{ArcTan}[p] - \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2 - 4ap^2}}\right]\right) + \frac{(a+1)*p}{\left((1+a + \sqrt{(-1+a)^2 - 4ap^2})\right)} - (w + b * \cos w) / (b^2 - 1)\right] - \frac{(-1+a)^2 \pi}{8a}$$

When $a < 1$

$$b := (a+1) / (a-1)$$

$$w := \text{ArcTan}[p] - \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2 - 4ap^2}}\right] - \pi / 2$$

Finding $\sin(w)$

$$\text{FullSimplify}[\text{TrigExpand}[\text{Sin}[w]]]$$

$$\frac{-((1+a)p^2) - \sqrt{(-1+a)^2 - 4ap^2}}{\sqrt{1+p^2} \sqrt{\frac{(-1+a)^2 (1+p^2)}{(-1+a)^2 - 4ap^2}} \sqrt{(-1+a)^2 - 4ap^2}}$$

Finding $\cos(w)$

$$\text{FullSimplify}[\text{TrigExpand}[\text{Cos}[w]]]$$

$$\frac{p(-1-a + \sqrt{(-1+a)^2 - 4ap^2})}{\sqrt{1+p^2} \sqrt{\frac{(-1+a)^2 (1+p^2)}{(-1+a)^2 - 4ap^2}} \sqrt{(-1+a)^2 - 4ap^2}}$$

$$\sin w := \left((1+a)p^2 + \sqrt{(-1+a)^2 - 4ap^2}\right) / ((a-1) * (1+p^2))$$

$$\cos w := \left(p(1+a - \sqrt{(-1+a)^2 - 4ap^2})\right) / ((a-1) * (1+p^2))$$

Checking (34)

$$\text{FullSimplify}\left[-\text{Log}\left[1+a + \sqrt{(-1+a)^2 - 4ap^2}\right] - \frac{(a+1)}{\left(1+a + \sqrt{(-1+a)^2 - 4ap^2}\right)} - ((b^2 - 1) * \text{Log}[b - \sin w] + b * \sin w) / (b^2 - 1)\right] - \frac{(1+a)^2}{4a} - \text{Log}\left[\frac{1+a - \sqrt{(-1+a)^2 - 4ap^2}}{(-1+a)(1+p^2)}\right] - \text{Log}\left[1+a + \sqrt{(-1+a)^2 - 4ap^2}\right]$$

$$\text{FullSimplify}\left[\left(1+a+\sqrt{(-1+a)^2-4ap^2}\right)\left(1+a-\sqrt{(-1+a)^2-4ap^2}\right)\right]/\left((-1+a)(1+p^2)\right)\right]$$

$$\frac{4a}{-1+a}$$

Thus it is

$$-\frac{(1+a)^2}{4a} - \text{Log}\left[\text{ABS}\left[\frac{4a}{-1+a}\right]\right]$$

$$-\frac{(1+a)^2}{4a} - \text{Log}\left[\text{ABS}\left[\frac{4a}{-1+a}\right]\right]$$

Checking (35)

$$\text{FullSimplify}\left[(a-1)^2/(4a) * \left(\text{ArcTan}[p] - \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2-4ap^2}}\right]\right) + \right.$$

$$\left. (a+1) * p / \left(\left(1+a+\sqrt{(-1+a)^2-4ap^2}\right) - (w+b*\cos w)/(b^2-1)\right) \right]$$

$$\frac{(-1+a)^2 \pi}{8a}$$

Case “+“ between tangents.

$$b := (a+1)/(a-1)$$

$$w := \text{ArcTan}[p] + \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2-4ap^2}}\right] - \text{Pi}/2 * \text{Sign}[a-1]$$

When $a > 1$

$$w := \text{ArcTan}[p] + \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2-4ap^2}}\right] - \text{Pi}/2$$

Finding $\sin(w)$

$$\text{FullSimplify}[\text{TrigExpand}[\text{Sin}[w]]]$$

$$\frac{(1+a)p^2 - \sqrt{(-1+a)^2-4ap^2}}{\sqrt{1+p^2} \sqrt{\frac{(-1+a)^2(1+p^2)}{(-1+a)^2-4ap^2}} \sqrt{(-1+a)^2-4ap^2}}$$

Finding $\cos(w)$

FullSimplify[TrigExpand[Cos[w]]]

$$\frac{p \left(1 + a + \sqrt{(-1 + a)^2 - 4 a p^2}\right)}{\sqrt{1 + p^2} \sqrt{\frac{(-1 + a)^2 (1 + p^2)}{(-1 + a)^2 - 4 a p^2}} \sqrt{(-1 + a)^2 - 4 a p^2}}$$

$$\sin w := \left((1 + a) p^2 - \sqrt{(-1 + a)^2 - 4 a p^2} \right) / ((a - 1) * (1 + p^2))$$

$$\cos w := \left(p \left(1 + a + \sqrt{(-1 + a)^2 - 4 a p^2}\right) \right) / ((a - 1) * (1 + p^2))$$

Checking (34)

$$\begin{aligned} & \text{FullSimplify} \left[-\text{Log} \left[\left(1 + a - \sqrt{(-1 + a)^2 - 4 a p^2}\right) \right] - \right. \\ & \quad \left. (a + 1) / \left(1 + a - \sqrt{(-1 + a)^2 - 4 a p^2}\right) - ((b^2 - 1) * \text{Log}[b - \sin w] + b * \sin w) / (b^2 - 1) \right] \\ & - \frac{(1 + a)^2}{4 a} - \text{Log} \left[1 + a - \sqrt{(-1 + a)^2 - 4 a p^2} \right] - \text{Log} \left[\frac{1 + a + \sqrt{(-1 + a)^2 - 4 a p^2}}{(-1 + a) (1 + p^2)} \right] \end{aligned}$$

$$\begin{aligned} & \text{FullSimplify} \left[\left(1 + a + \sqrt{(-1 + a)^2 - 4 a p^2}\right) * \left(1 + a - \sqrt{(-1 + a)^2 - 4 a p^2}\right) / ((-1 + a) (1 + p^2)) \right] \\ & \frac{4 a}{-1 + a} \end{aligned}$$

Thus it is

$$\begin{aligned} & - \frac{(1 + a)^2}{4 a} - \text{Log} \left[\frac{4 a}{-1 + a} \right] \\ & - \frac{(1 + a)^2}{4 a} - \text{Log} \left[\frac{4 a}{-1 + a} \right] \end{aligned}$$

Checking (35)

$$\begin{aligned} & \text{FullSimplify} \left[(a - 1)^2 / (4 a) * \left(\text{ArcTan}[p] + \text{ArcTan} \left[\frac{(1 + a) p}{\sqrt{(-1 + a)^2 - 4 a p^2}} \right] \right) + \right. \\ & \quad \left. (a + 1) * p / \left(\left(1 + a - \sqrt{(-1 + a)^2 - 4 a p^2}\right) \right) - (w + b * \cos w) / (b^2 - 1) \right] \\ & \frac{(-1 + a)^2 \pi}{8 a} \end{aligned}$$

When $a < 1$

$$b := (a + 1) / (a - 1)$$

$$w := \text{ArcTan}[p] + \text{ArcTan} \left[\frac{(1 + a) p}{\sqrt{(-1 + a)^2 - 4 a p^2}} \right] + \text{Pi} / 2$$

Finding sin(w)

FullSimplify[**TrigExpand**[**Sin**[w]]]

$$\frac{-\left((1+a)p^2\right) + \sqrt{(-1+a)^2 - 4ap^2}}{\sqrt{1+p^2} \sqrt{\frac{(-1+a)^2(1+p^2)}{(-1+a)^2 - 4ap^2}} \sqrt{(-1+a)^2 - 4ap^2}}$$

Finding cos(w)

FullSimplify[**TrigExpand**[**Cos**[w]]]

$$-\frac{p\left(1+a+\sqrt{(-1+a)^2-4ap^2}\right)}{\sqrt{1+p^2} \sqrt{\frac{(-1+a)^2(1+p^2)}{(-1+a)^2-4ap^2}} \sqrt{(-1+a)^2-4ap^2}}$$

$$\sin w := \left((1+a)p^2 - \sqrt{(-1+a)^2 - 4ap^2}\right) / ((a-1) * (1+p^2))$$

$$\cos w := \left(p\left(1+a+\sqrt{(-1+a)^2 - 4ap^2}\right)\right) / ((a-1) * (1+p^2))$$

Checking (34)

$$\begin{aligned} & \text{FullSimplify}\left[-\text{Log}\left[\left(1+a-\sqrt{(-1+a)^2-4ap^2}\right)\right] - \right. \\ & \quad \left. (a+1) / \left(1+a-\sqrt{(-1+a)^2-4ap^2}\right) - ((b^2-1) * \text{Log}[b-\sin w] + b * \sin w) / (b^2-1)\right] \\ & - \frac{(1+a)^2}{4a} - \text{Log}\left[1+a-\sqrt{(-1+a)^2-4ap^2}\right] - \text{Log}\left[\frac{1+a+\sqrt{(-1+a)^2-4ap^2}}{(-1+a)(1+p^2)}\right] \end{aligned}$$

$$\begin{aligned} & \text{FullSimplify}\left[\left(1+a+\sqrt{(-1+a)^2-4ap^2}\right) * \left(1+a-\sqrt{(-1+a)^2-4ap^2}\right) / ((-1+a)(1+p^2))\right] \\ & \frac{4a}{-1+a} \end{aligned}$$

Thus it is

$$\begin{aligned} & -\frac{(1+a)^2}{4a} - \text{Log}\left[\text{ABS}\left[\frac{4a}{-1+a}\right]\right] \\ & -\frac{(1+a)^2}{4a} - \text{Log}\left[\text{ABS}\left[\frac{4a}{-1+a}\right]\right] \end{aligned}$$

Checking (35)

$$\begin{aligned} & \text{FullSimplify}\left[(a-1)^2 / (4a) * \left(\text{ArcTan}[p] + \text{ArcTan}\left[\frac{(1+a)p}{\sqrt{(-1+a)^2-4ap^2}}\right]\right) + \right. \\ & \quad \left. (a+1) * p / \left(\left(1+a-\sqrt{(-1+a)^2-4ap^2}\right)\right) - (w + b * \cos w) / (b^2-1)\right] \\ & - \frac{(-1+a)^2 \pi}{8a} \end{aligned}$$

2.4 Derivation of the determinant of (39)

$$a * (1 + g'[v]^2) * X^2 + (a + 1) * \text{Sqrt}[g''[v]] * X * Y + g''[v] * Y^2 - (a + 1) * \text{Sqrt}[g''[v]] * g'[v] * X + g''[v] == 0$$

$$a X^2 (1 + g'[v]^2) + (1 + a) X Y \sqrt{g''[v]} - (1 + a) X g'[v] \sqrt{g''[v]} + g''[v] + Y^2 g''[v] == 0$$

$$\text{FullSimplify[Det[{{a * (1 + g'[v]^2), (a + 1) * Sqrt[g''[v]] / 2, -(a + 1) * Sqrt[g''[v]] * g'[v] / 2}, \{(a + 1) * Sqrt[g''[v]] / 2, g''[v], 0\}, \{-(a + 1) * Sqrt[g''[v]] * g'[v] / 2, 0, g''[v]\}}]]]$$

$$-\frac{1}{4} (-1 + a)^2 (1 + g'[v]^2) g''[v]^2$$

2.5 Checking of (44)

$$\text{FullSimplify}\left[\partial_u \left(\frac{-f'[u] g''[v] + h''[v]}{f''[u]} \right) - (g''[v] * (f'''[u] * f'[u] - f''[u]^2) - h'''[v] * f'''[u]) / f''[u]^2\right]$$

$$0$$

2.6 Checking of (50)

$$L[Y_] := g''[v] - h''[v] * (g'[v] - Y) / h'[v]$$

$$\text{FullSimplify[Discriminant[a * ((1 + g'[v]^2) * X + (Y^2 + 1) * L[Y])^2 - (a + 1)^2 * X * (Y - g'[v])^2 * L[Y], X] - (a + 1)^2 * (Y - g'[v])^2 * L[Y]^2 * ((a - 1)^2 * g'[v]^2 - 4 * a - 2 * (a + 1)^2 * g'[v] * Y + ((a - 1)^2 - 4 * a * g'[v]^2) * Y^2)]]$$

$$0$$

2.7 Derivation of the discriminant of the square free term of (50)

$$\text{FullSimplify[Discriminant[(a - 1)^2 * g'[v]^2 - 4 * a - 2 * (a + 1)^2 * g'[v] * Y + ((a - 1)^2 - 4 * a * g'[v]^2) * Y^2, Y]]]$$

$$16 (-1 + a)^2 a (1 + g'[v]^2)^2$$

3. Euclidean rotational CRPC surfaces

```
Simplify[
  D[Hypergeometric2F1[1/2, 1/2 + 1/(2a), 3/2 + 1/(2a), r^(2a)] * r^(a+1)/(a+1), r] -
  r^a/Sqrt[1 - r^(2a)]]
0
```

4. Dual – translational surfaces

Here we provide the derivation of (53) and (54).

4.1. Derivation of (53)

```
x[u_, v_] := v + b * Cos[v]
y[u_, v_] := b * Sin[v] + (b * b - 1) * Log[b - Sin[v]] + (1 - b^2) * u
z[u_, v_] := Exp[u]

Solve[∂uz[u, v] == zx * ∂ux[u, v] + zy * ∂uy[u, v] &&
  ∂vz[u, v] == zx * ∂vx[u, v] + zy * ∂vy[u, v], {zx, zy}]
```

$$\left\{ \left\{ zx \rightarrow \frac{e^u \cos[v]}{(-1 + b^2)(b - \sin[v])}, zy \rightarrow -\frac{e^u}{-1 + b^2} \right\} \right\}$$

$$\{zx, zy\} = \left\{ \frac{e^u \cos[v]}{(-1 + b^2)(b - \sin[v])}, -\frac{e^u}{-1 + b^2} \right\}$$

$$\left\{ \frac{e^u \cos[v]}{(-1 + b^2)(b - \sin[v])}, -\frac{e^u}{-1 + b^2} \right\}$$

```
zstar[u_, v_] := FullSimplify[x[u, v] * zx + y[u, v] * zy - z[u, v]]
zstar[u, v]
```

$$e^u \left(-1 + u - \log[b - \sin[v]] + \frac{b + v \cos[v] - b^2 \sin[v]}{(-1 + b^2)(b - \sin[v])} \right)$$

4.2. Derivation of (54)

```
x[u_, v_] := u + v
y[u_, v_] := Log[Cos[u]] - Log[Cos[v]]
z[u_, v_] := u

Solve[∂uz[u, v] == zx * ∂ux[u, v] + zy * ∂uy[u, v] &&
  ∂vz[u, v] == zx * ∂vx[u, v] + zy * ∂vy[u, v], {zx, zy}]
```

$$\left\{ \left\{ zx \rightarrow \frac{\tan[v]}{\tan[u] + \tan[v]}, zy \rightarrow -\frac{\cot[u]}{1 + \cot[u] \tan[v]} \right\} \right\}$$

$$\{zx, zy\} = \left\{ \frac{\tan[v]}{\tan[u] + \tan[v]}, -\frac{\cot[u]}{1 + \cot[u] \tan[v]} \right\}$$

$$\left\{ \frac{\tan[v]}{\tan[u] + \tan[v]}, -\frac{\cot[u]}{1 + \cot[u] \tan[v]} \right\}$$

zstar[u_, v_] := FullSimplify[x[u, v] * zx + y[u, v] * zy - z[u, v]]

zstar[u, v]

$$\frac{-\log[\cos[u]] + \log[\cos[v]] - u \tan[u] + v \tan[v]}{\tan[u] + \tan[v]}$$